



Production and MRP: Forecast Consumption

SAP Business One
Version 10.0

PUBLIC

THE BEST RUN 

Welcome to the topic on consuming forecasts in Material Requirements Planning (MRP). This topic builds on what you have learned in the Introduction to MRP topic.

Objectives



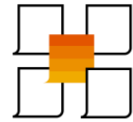
At the end of this topic, you will be able to:

- Configure a consumption method for an MRP forecast
- Explain how MRP results are affected when forecasts are consumed

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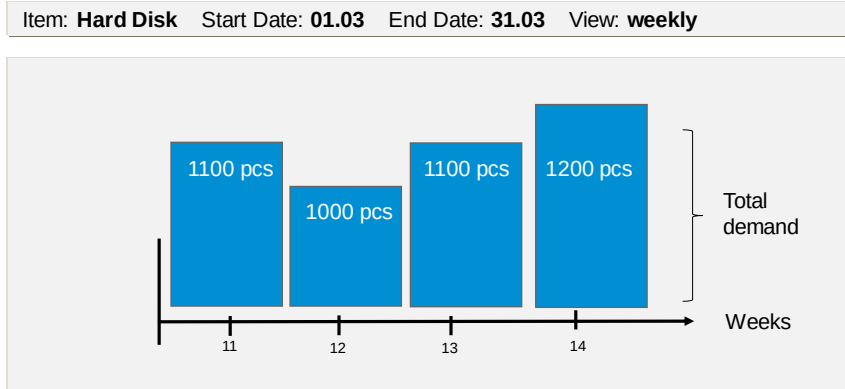
Business Scenario



- OEC Computers uses MRP to plan their sales inventory
- Previously when we they ran MRP, they had problems with backorders for some items and overstocks on others
- Consuming forecasts will enable them to more accurately manage their inventory and save money by purchasing wisely

- OEC Computers uses MRP to plan their inventory for the office equipment and supplies that they sell
- Previously when we they ran MRP, they had problems with backorders for some items and overstocks on others
- Consuming forecasts will enable them to more accurately manage their inventory and save money by purchasing wisely

Forecast Concept



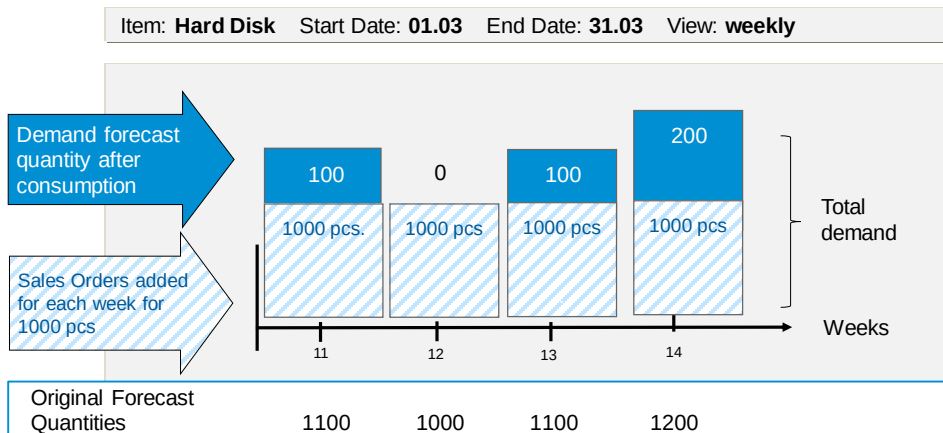
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Before we dive into forecast consumption rules let us go over the main forecast principles as described in the MRP Process course topic.

- You can create forecasts to plan purchasing, production or transfers in advance, even before you receive other actual requirements like sales orders.
- By using a forecast, you can purchase or produce items according to the forecast demand. When the actual sales orders arrive, you are able to supply the goods even at short notice.
- In the image we see an illustration of a weekly forecast for the hard disk item. Each week has a different forecasted quantity.

Forecast Consumption Concept



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Now let's look at how a forecast is consumed.

In the previous topic we discussed how that the forecast appears on the demand row of the MRP report, much like the sales order. However, when the purpose of the forecast is to anticipate sales orders before they arrive, we would like to subtract the actual sales order quantity from the forecast to avoid duplication of demand.

SAP Business One gives you the option to set sales orders and blanket agreements either to increase demand in addition to your forecast or to consume forecasted quantities to avoid the duplication of the demand. When you set sales orders to consume forecast, the system subtracts the sales order quantity from the forecasted quantity if there is a demand from both a forecast and sales orders during a certain period. Then the system displays the remaining quantity from the forecast in addition to the sales order quantity. If a forecast is fully consumed, it no longer appears in the MRP report.

Let's look at an example where we have set the sales order to consume forecast. In the image we see an illustration of a forecast consumption. In this example, each week contains sales orders for 1000 pieces. We can see that the sales orders consume the forecast of each week. The quantity left (in blue) will be displayed as the forecast demand in the MRP report.

In the demand row of the MRP report, we will see a total demand of **1100** units in week 11, **1000** in week 12, **1100** in week 13 and **1200** in week 14.

Note that the consumption does not change the actual forecast defined, only the demand displayed in the MRP report. If the sales orders were for larger quantities than the forecast, additional quantity would be added to the demand.

Consume Forecast Settings

You can activate forecast consumption at the company level

General Settings

BP Budget Services Display Font & Bkgd Path **Inventory**

Items **Planning** Reporting

☒ Consume Forecast

Consumption Method Backward-Forward

Days Backward 7 Backward-Forward

Days Forward 7 Forward-Backward

General Settings → Inventory → Planning

- Company level settings default for new sales orders and blanket agreements
- Consume Forecast option can be turned on/off at row level

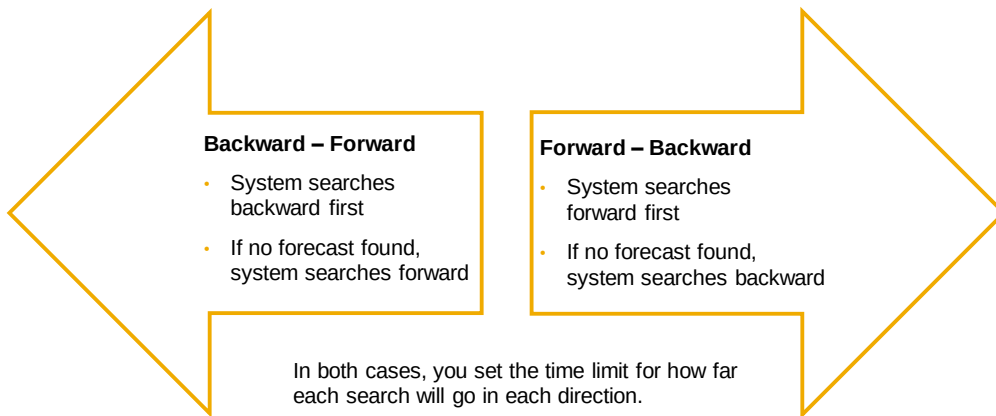
Sales Order

Contents Logistics Accounting Attachments

#	Item No.	Item Description	Quantity	Whse	Unit Price	Total (LC)	Consume Forecast
1	C00007	Hard Disk Seagate 400 GB	90	01	GBP 120.00	GBP 10,800.00	Yes
2	C00009	Keyboard Comfort USB	120	01	GBP 15.00	GBP 1,800.00	Yes
3							Yes

- You can activate forecast consumption at the company level, but you can also control consumption for each row in the sales order. The same is true for sales blanket agreements and we will cover this later in the course.
- In the *Planning* tab of the *General Settings* window you can define whether forecasts are to be consumed by sales orders. This will be the default for all new sales orders. If you wish to prevent the consumption of a certain sales order, choose *No* in the *Consume Forecast* column of the sales order.
- After you have decided to consume a forecast, how does the system know which forecast to consume? Is it the forecast you defined for the week before the due date of the sales order or maybe it is the forecast of the week after? This is controlled by the *Consumption Method* and the *Days Backward* and *Days Forward* parameters.
- We will now explain how these definitions affect forecast consumption by using two scenarios.

Two Consumption Methods



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When you've set up sales orders and/or blanket agreements to consume forecasts, the system looks to see if any relevant documents are included in the MRP run. When for example, a sales order is included, the MRP wizard then takes the quantities for the item(s) into account and searches for existing forecasts that are closest to the sales order delivery date. This search is controlled by the consumption method assigned in the General Settings.

There are two consumption methods: Backward-Forward and Forward-Backward.

Using Backward-Forward, the system searches backward from the delivery date of the sales for a forecast for the number of days you have indicated. If no forecast is found, the system then searches forward from the delivery date within the range set. In this way, the Backward-Forward method consumes the forecast defined for earlier dates first and therefore tends to minimize demand in the near future which helps keep down the cost of inventory.

Using Forward-Backward, the system searches in the opposite manner. The system searches ahead starting from the sales order delivery date up to the limit of days you have set. If no forecast is found, the system then searches backwards starting from the sales order delivery date. In this way, Backward-Forward increases near future recommendations and more likely increases inventory. This setting is useful if you are concerned about minimizing a possible

risk of unfulfilled demand.

In both cases, you set the time limit for how far each search will go in each direction.

Example Scenarios

We will look at 3 scenarios:

1. Without consuming forecasts
2. Using Backward-Forward Consumption Method
 - Limit on days searched backward: 6
 - Limit on days searched forward: 6
3. Using Forward-Backward Consumption Method
 - Limit on days searched forward: 6
 - Limit on days searched backward : 6

We will look at 3 scenarios:

1. Without consuming forecasts
2. Using Backward-Forward Consumption Method
 - Limit on days searched backward: 6
 - Limit on days searched forward: 6
3. Using Forward-Backward Consumption Method
 - Limit on days searched forward: 6
 - Limit on days searched backward : 6

Scenario for all examples

	Week 15							Week 16						
	M 8	T 9	W 10	Th 11	F 12	S 13	Su 14	M 15	T 16	W 17	Th 18	F 19	S 20	Su 21
Sales Order Quantities	200				900			300						
Weekly Forecast	1000							1000						
Total Demand														

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For comparing the effects of each method, we will assume the same quantities in each example.

There are always 3 sales orders coming in. Each sales order has a different delivery date. The first two sales orders fall within week 15 on Monday the 8th and on Friday the 12th. The third sales order is on Monday the 15th so it falls in the week 16.

We will assume a sales forecast has been defined with a weekly quantity of 1000. Since the forecast is defined weekly, the quantity is assigned to the first day of the week and in our scenario it is Monday. (Periodic forecasts are always assigned to the first day of the week.)

Example 1: Without consuming forecasts

	Week 15							Week 16						
	M	T	W	Th	F	S	Su	M	T	W	Th	F	S	Su
	8	9	10	11	12	13	14	15	16	17	18	19	20	21
Sales Order Quantities	200				900			300						
Weekly Forecast	1000							1000						
Total Demand	2100	= 1000 + 200 + 900						1300	= 1000 + 300					

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If we choose not to use a consumption method, then the total demand would be the sales orders quantity plus the forecasted quantity.

In that case the total demand for week 15 would be **2100** and for week 16 would be **1300**.

Example 2: Backward-Forward Consumption Method (1 of 4)

6 days backward, 6 days forward

	Week 15							Week 16						
	M	T	W	Th	F	S	Su	M	T	W	Th	F	S	Su
	8	9	10	11	12	13	14	15	16	17	18	19	20	21
Sales Order Quantities	200				900			300						
Weekly Forecast	1000 800							1000						
Total Demand														

First sales order consumes 200 of 1000 in forecast

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In example 2, we use the configuration settings:

- Backward Forward Consumption Method
- 6 days backward
- 6 days forward

For each sales order, the system will look 6 days backward and if no unconsumed forecast quantity is found, it will look 6 days forward.

We will examine the effect of each sales order in a series of 4 slides.

The first sales order on Monday the 8th has a quantity of 200. Since this is the same day of the forecasted weekly quantity, the sales order consumes 200 of the 1000.

Example 2: Backward-Forward Consumption Method (2 of 4)

6 days backward, 6 days forward

	Week 15							Week 16						
	M	T	W	Th	F	S	Su	M	T	W	Th	F	S	Su
	8	9	10	11	12	13	14	15	16	17	18	19	20	21
Sales Order Quantities	200				900			300						
Weekly Forecast	1000							1000						
	800							900						
	0													
Total Demand														

The second sales order looks back to find a forecast, finds one 4 days back and consumes the remaining 800 of forecast

No additional forecast quantity is available in the past, so system looks forward. Second sales order consumes 100 of Week 16 forecast

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The second sales order is for 900.

The system looks backward to find a forecast with quantity to consume.

The Week 15 forecast has a remaining quantity of 800. The second sales order consumes the entire quantity, but still needs to consume 100.

The system looks back 6 days from Friday the 12th to Saturday the 6th, but there is no additional forecast quantity available.

Since no additional forecast quantity is available in the past, the system looks forward and finds Week 16 forecast.

The second sales order then consumes a quantity of 100 of the Week 16 forecast.

Notice that consumption is not always taken from the same period. If the first period searched does not have forecasted quantity to consume, the system can consume quantity in a different period.

Example 2: Backward-Forward Consumption Method (3 of 4)

6 days backward, 6 days forward

	Week 15							Week 16						
	M	T	W	Th	F	S	Su	M	T	W	Th	F	S	Su
	8	9	10	11	12	13	14	15	16	17	18	19	20	21
Sales Order Quantities	200				900			300	The third sales order consumes 300 from Week 16.					
Weekly Forecast	1000 800 0							1000 900 600						
Total Demand														

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The third sales order is at the beginning of week 16. It consumes 300 of the 900 remaining in the weekly forecast on the same day.

This leaves a total of 600 in the weekly forecast.

Example 2: Backward-Forward Consumption Method (4 of 4)

6 days backward, 6 days forward

	Week 15							Week 16						
	M 8	T 9	W 10	Th 11	F 12	S 13	Su 14	M 15	T 16	W 17	Th 18	F 19	S 20	Su 21
Sales Order Quantities	200				900			300						
Weekly Forecast	1000 800 0							1000 900 600						
Total Demand	1100	= 200 + 900						900	= 300 + 600					

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The total demand for Week 15 is 1100. Since the forecast was fully consumed, it contains the quantity of the sales orders in that week.

The total demand for Week 6 is 900. The total demand equals the sales order quantity of 300 plus the remaining forecasted quantity of 600.

Example 3: Forward-Backward Consumption Method (1 of 4)

6 days forward, 6 days backward

	Week 15							Week 16						
	M	T	W	Th	F	S	Su	M	T	W	Th	F	S	Su
	8	9	10	11	12	13	14	15	16	17	18	19	20	21
Sales Order Quantities	200				900			300						
Weekly Forecast	1000 800							1000						
Total Demand														

First sales order consumes 200 of Week 15 forecast

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Now we will look at the second Consumption Method option: Forward-Backward. We have kept the parameters of days the same: 6 days forward and 6 days backward.

The first sales order (for Monday the 8th), once again consumes 200 units of the weekly forecast in Week 15. This leaves the forecast with a total of 800.

Example 3: Forward-Backward Consumption Method (2 of 4)

6 days forward, 6 days backward

	Week 15							Week 16						
	M	T	W	Th	F	S	Su	M	T	W	Th	F	S	Su
	8	9	10	11	12	13	14	15	16	17	18	19	20	21
Sales Order Quantities	200				900			300						
Weekly Forecast	1000							1000						
	800							100						
Total Demand														

The second sales order looks forward and finds the Week 16 forecast 3 days ahead.

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The second sales order (from Friday the 12th) looks forward and finds the forecast of week 16, which is 3 days ahead.

It consumes 900 units from the Week 16 forecast.

Example 3: Forward-Backward Consumption Method (3 of 4)

6 days forward, 6 days backward

	Week 15							Week 16						
	M	T	W	Th	F	S	Su	M	T	W	Th	F	S	Su
	8	9	10	11	12	13	14	15	16	17	18	19	20	21
Sales Order Quantities	200				900			300						
Weekly Forecast	1000 800							1000 100						
Total Demand														

The third sales order consumes the remaining quantity of the Week 16 forecast.

The third sales order (from Monday the 15th) consumes the remaining 100 of the Week 16 forecast.

Example 3: Forward-Backward Consumption Method (4 of 4)

6 days forward, 6 days backward

	Week 15							Week 16						
	M 8	T 9	W 10	Th 11	F 12	S 13	Su 14	M 15	T 16	W 17	Th 18	F 19	S 20	Su 21
Sales Order Quantities	200				900			300						
Weekly Forecast	1000 800							1000 100						
Total Demand	1900	= 200 + 900 + 800						300	All of forecast is consumed so demand is based on sales order qty					

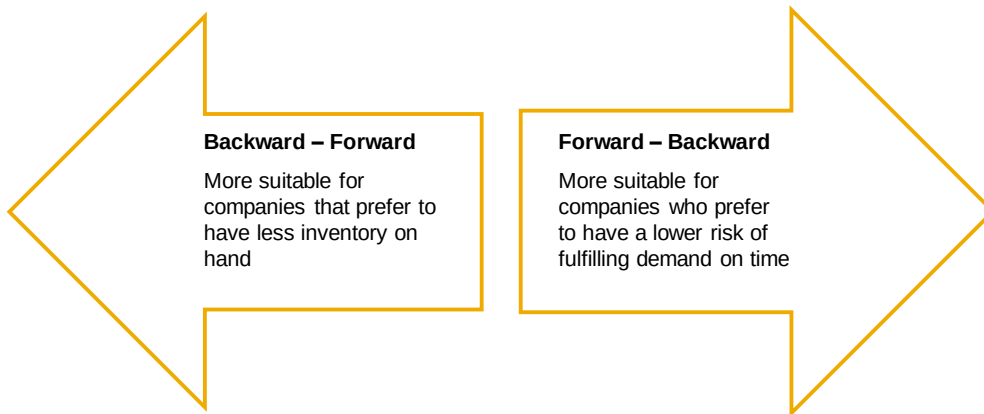
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With this method, we see that much of the demand has moved to Week 15 since demand is consumed from the future first.

- In week 15, the demand is 1900 units. This quantity is the total of the two sales orders for week 15 (200 + 900) plus what is left from the week 15 forecast (800)
- In week 16, the total demand is 300 units. All of the forecasted quantity has been consumed by the second and third sales orders.

Considerations for which method to choose



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- When choosing *Forecast Consumption Method*, it is important to consider the implications of each method and choose carefully.
- In our example scenario, the demand of each week varied when we used different consumption methods. If we sum the total demand of both weeks, we receive a different quantity in each method. In the *Forward-Backward* method the demand of both weeks is **2,200** (1,900+300) units and with the *Backward-Forward* method it is only **2,000** (1,100+900) units.
- When you come to decide which method to use, you should consider the following:
 - The *Backward-Forward* method is more suitable for companies that prefer to have less inventory on hand.
 - The *Forward-Backward* method is more suitable for companies who prefer to have a lower risk of fulfilling demand on time.
- In the *Backward-Forward* method you consume the forecast that is defined for the near future first and thus minimizing the demand of the near future. Lower demand leads to a lower recommendation quantity and lower purchase and production.
- The advantage of this method is the low inventory level remaining after the fulfilment of the demand.
- But it also means a higher risk of not fulfilling new demand on time.
- The *Forward-Backward* on the other hand increases the near future recommendations and thus raising inventory level and minimizing the risk of unfulfilled demand.

- Since you can change the *Forecast Consumption Method* at any point, you always have the option to rerun MRP and compare the results.

Blanket Agreements and Forecast Consumption

- Sales blanket agreements can consume forecasts similar to sales orders
- Company level settings control consumption in blanket agreements
- You can turn on/off consumption at the row level
- MRP considers blanket agreements that:
 - Have the status *Approved*
 - Contain a warehouse on the row

The image shows two SAP windows. The top window is 'Draft Sales Blanket Agreement' with fields for BP Code (C20000), BP Name (Maxi-Teq), Contact Person (Norm Thompson), Customer Ref. No., BP Currency (GBP), Telephone No. (020 5894 9487), E-Mail (norm.thompson@maxi-teq), Agreement Method (Primary), Items Method, Start Date (02.08.), End Date, BP Project, Termination Date, and Signing Date (02.08.). Below these are tabs for General, Details, Documents, Attachments, and Recurring Transactions. The bottom window is 'Row Details - Blanket Agreement' showing a table with columns: #, Frequency, From, To, Item No., Item Description, Item Group, Planned Quantity, Unit Price, Cumulative Commitment, and Warehouse. It contains two rows for 'Tablet PC 64G/PCs' with a frequency of 'Monthly' and a quantity of 2,100. The bottom row has a warehouse code '2,100' in the Warehouse column. Buttons for OK and Cancel are at the bottom.

#	Frequency	From	To	Item No.	Item Description	Item Group	Planned Quantity	Unit Price	Cumulative Commitment	Warehouse
1	Monthly	01.09.	31.12.	200001	Tablet		2,100			
2	Monthly			200001	Tablet					2,100

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Sales blanket agreements can consume forecasts much like sales orders.

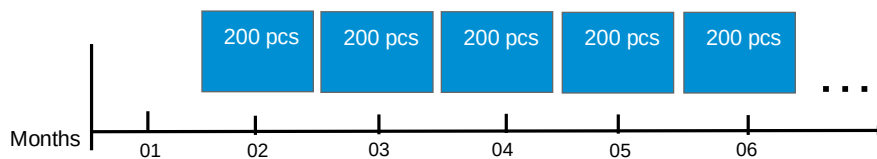
Just like sales orders, the company-wide setting in General Settings defaults into new documents.

You can turn on or off forecast consumption at the row level of blanket agreements

MRP only considers blanket agreements with the status *Approved* and with a warehouse indicated on the row.

Blanket Agreement Consumption Example

- OEC Computers has a blanket agreement with Maxi-Teq for 2100 units.
- Each month Maxi-Teq will order a fixed quantity of 200 units based on blanket agreement.
- The blanket agreement row *From Date* is February 1
- In order to calculate consumption, the system calculates the expected delivery date of future sales orders and thus sets the consumption date.



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- Here is an example of blanket agreement consumption:
 - OEC Computers has a blanket agreement with Maxi-Teq for 2100 units. Each month they order a fixed quantity of 200 units.
 - Metaphorically, you can think of a sales blanket agreement as similar to a customer-specific forecast that can be used to consume your regular sales forecast. It allows you to anticipate demand from the future sales orders from this customer.
 - In order to calculate consumption, the system calculates the expected sales orders delivery due date and thus sets the consumption date.
 - Frequencies in blanket agreement rows can be set for Daily, Weekly, Monthly, Quarterly, Semi-annually and One time.
 - No matter which frequency we choose in the *Frequency* field, the first consumption day equals the *From Date* in the row. In our example the first consumption of 200 units occurs on the *From Date* of February 1.
 - Then, every other consumption is set to the 1st of each of the following months.
 - If choosing a *One Time* occurrence, the consumption is also due on the *From Date* in the row, in this case February 1st.

Summary



Here are some key points to take away from this course:

- SAP Business One allows forecasted demand to be consumed by sales order and blanket agreement rows.
- You can activate forecast consumption at the company level or control it at the row level.
- There are two choices of consumption methods: Backward-Forward and Forward-Backward.
 - The *Backward-Forward* method is more suitable if you wish to reduce inventory on hand.
 - The *Forward-Backward* method is more suitable if you wish to lower the risk of fulfilling demand on time.
- The settings for consumption are set in the *Planning* window of the *Inventory* tab in *General Settings*.

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- SAP Business One allows forecasted demand to be consumed by sales order and blanket agreement rows.
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- There are two choices of consumption methods: Backward-Forward and Forward-Backward.
 - The *Backward-Forward* method is more suitable if you wish to reduce inventory on hand.
 - The *Forward-Backward* method is more suitable if you wish to lower the risk of fulfilling demand on time.
- The settings for consumption are set at the company level in the *Planning* window of the *Inventory* Tab in *General Settings*. In this window, you set the consumption method and the number of days backward and the number of days forward that the system will look for a forecast.

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